AI Powered Resume Builder/Analyzer

A SYNOPSIS SUBMITTED

To

SAVITRIBAI PHULE PUNE UNIVERSITY

FOR THE PARTIAL FULFILMENT OF AWARD OF DEGREE

BACHELOR OF ENGINEERING
In

Computer Engineering

By

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UNDER THE GUIDANCE OF
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DEPARTMENT OF COMPUTER ENGINEERING

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Academic Year: 2025-26

SYNOPSIS

1. Project Group Information

Group Id: 01
Name of Students:

- Phadtare Akash Ashok
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- Jadhav Ganesh Motiram
- Jadhav Shubham Dattatray
- Kadam Shrikant Nagnath

2. Project Title

AI Powered Resume Builder/Analyzer

3. Project Sponsorship details

Sponsorship Company: Dc Tech core Private Limited
Sponsoring Company Address: Office No.108, Siddhivinayak Angan, Navale Brg, near Me, nearby Airtel thanks office, Khedekar Nagar, Narhe, Pune, Maharashtra 411041

4. Problem Statement

Job seekers struggle to create ATS-friendly resumes and tailor them to job descriptions, leading to lower chances of shortlisting.

5. Area of Project

The project lies in the domain of Artificial Intelligence, Natural Language Processing (NLP), and Web Application Development.

6. Abstract

In today's competitive job market, the resume is the first and most critical tool that connects job seekers with potential employers. However, a significant challenge faced by applicants is the inability to design resumes that are both professionally structured and optimized for Applicant Tracking Systems (ATS). Many candidates fail to align their resumes with specific job descriptions, resulting in low shortlisting rates despite having relevant skills and experience. To address this issue, this project proposes the development of an **AI-powered Resume Analyzer and Builder web application** that leverages Artificial Intelligence and Natural Language Processing (NLP) to create, evaluate, and optimize resumes.

The proposed system consists of two main modules: **Resume Analyzer** and **Resume Builder**. In the Resume Analyzer, users can upload their resumes in PDF or DOCX format along with a job description. The system extracts relevant information such as education, experience, skills, and certifications using NLP-based text parsing techniques. It then performs keyword matching, evaluates ATS compatibility, and generates a comprehensive scorecard highlighting strengths, weaknesses, and missing skills. Suggestions are provided to improve content, grammar, structure, and keyword alignment to enhance the resume's chances of passing through automated screening systems.

The Resume Builder module enables users to create new resumes from scratch using a step-by-step guided form. It incorporates **AI-assisted text generation** that suggests professional phrasing, impactful summaries, and industry-relevant skills. Multiple ATS-friendly templates are offered to ensure compatibility with hiring systems while maintaining a visually appealing layout. The system also allows real-time preview and easy export to PDF or DOCX formats, ensuring accessibility and convenience for end users.

From a technological perspective, the web application is designed using **modern web frameworks** for scalability and responsiveness. The frontend is developed with React/Next.js and Tailwind CSS for an interactive user experience, while the backend is powered by Node.js or Django/Flask for robust API integration. The AI layer employs NLP libraries such as spaCy, NLTK, and transformer-based models (BERT, GPT-based APIs) for text analysis, keyword extraction, and content generation. Data management is handled using MongoDB or PostgreSQL, while file storage is supported through cloud solutions.

This project not only simplifies the resume-building process but also provides intelligent insights for applicants to align their profiles with job market demands. By bridging the gap between job seekers and employer expectations, the system enhances employability and significantly increases the likelihood of successful shortlisting. Ultimately, the AI Resume Analyzer and Builder web application demonstrates the potential of combining Artificial Intelligence, NLP, and web technologies to create innovative solutions in the field of career development and recruitment technology.

System Flow Diagram

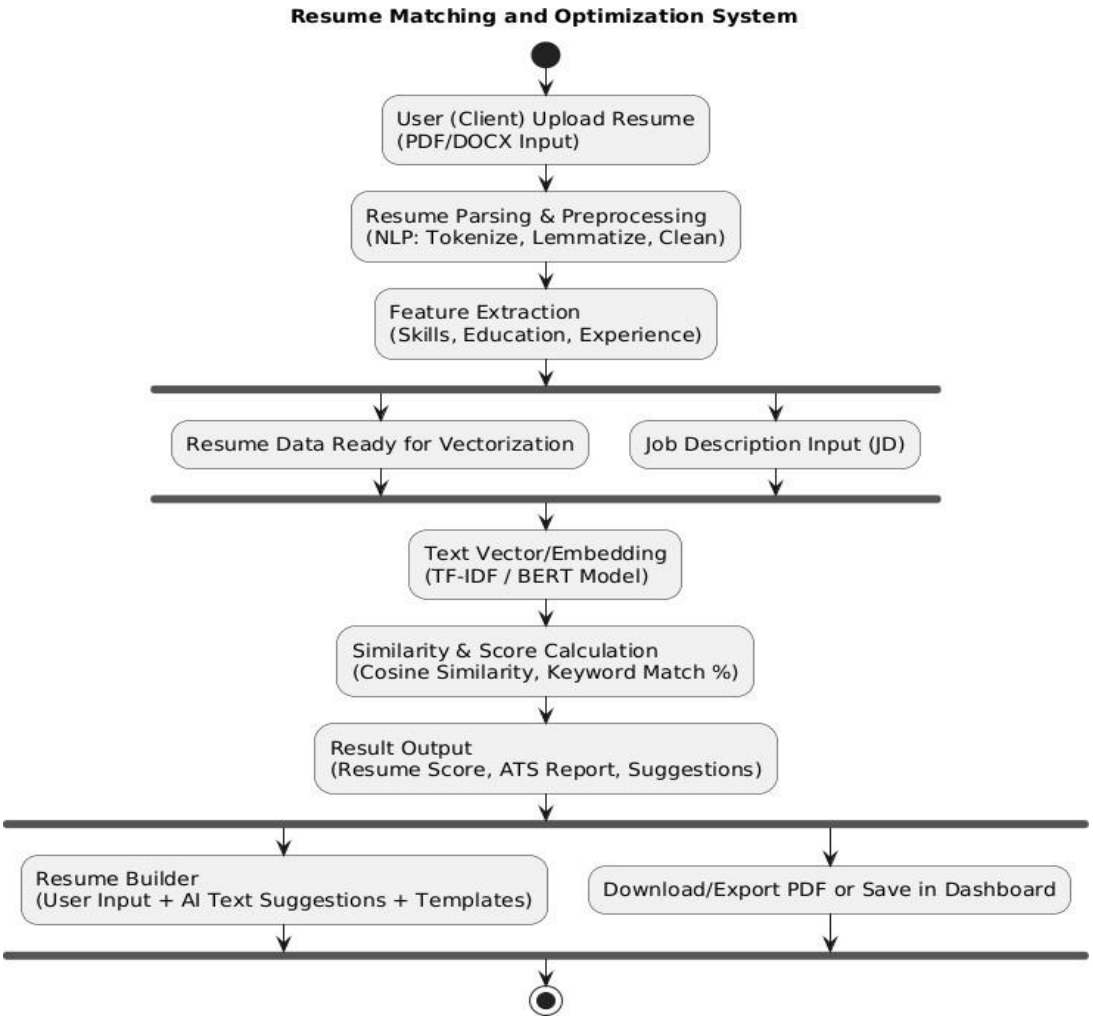


Figure 1: System Flow Diagram for AI Powered Resume Builder/Analyzer

7. Goals and Objectives

Goals

1. To develop an intelligent web application that assists job seekers in building professional and ATS-friendly resumes.
2. To leverage Artificial Intelligence and Natural Language Processing for accurate resume analysis and job description matching.
3. To enhance employability by providing personalized recommendations for improving resume quality and relevance.
4. To integrate modern web technologies for creating a scalable, user-friendly, and interactive platform.

Objectives

1. To design a Resume Analyzer module that evaluates resumes based on ATS compliance, keyword match, and formatting.
2. To implement a Resume Builder module that offers customizable templates and AI-powered content suggestions.
3. To enable users to upload resumes in PDF/DOCX format and generate a detailed performance scorecard.
4. To provide real-time feedback on missing skills, grammatical errors, and structure improvements.
5. To incorporate AI/NLP models (e.g., BERT, spaCy) for text parsing, keyword extraction, and professional wording.
6. To ensure easy resume export in standard formats (PDF/DOCX) for professional usage.
7. To build a secure and scalable web application with modern technologies (React, Node.js/Django, MongoDB/PostgreSQL).

8. Relevant mathematics associated with the Project:

The project employs several mathematical and statistical techniques to analyze, process, and score resumes against job descriptions. The core areas of mathematics include:

1. Linear Algebra & Vector Space Models
    - Resumes and job descriptions are converted into vector representations.
    - Techniques like TF-IDF (Term Frequency–Inverse Document Frequency) and word embeddings (Word2Vec, BERT embeddings) rely on vector algebra.
    - Example: Cosine Similarity is used to measure the closeness between resume vectors and job description vectors.
2. Probability & Statistics
    - Probabilistic models like Naïve Bayes may be used for text classification (skill identification, section classification).
    - Statistical frequency analysis is applied to identify keyword importance and missing skills.
3. Optimization Techniques
    - Machine learning models optimize weights during training to minimize loss functions.
    - Example: Cross-Entropy Loss for classification tasks in NLP.
4. Matrix Operations in NLP Models
    - Word embeddings and deep learning models like BERT use matrix multiplications, dot products, and transformations in multiple layers of neural networks.
    - Attention mechanisms in transformers compute weighted importance using Softmax:
5. Scoring Mechanism (Resume-Job Match)
    - The resume score is calculated using a weighted formula combining cosine similarity, keyword match ratio, and ATS compliance metrics.

Methodology

The methodology for developing the AI Resume Analyzer/Builder involves several phases, combining requirement analysis, system design, AI model integration, and web deployment. The steps are as follows:

1. Requirement Analysis

- Identify the problems faced by job seekers in creating ATS-friendly resumes.
- Gather functional requirements (resume analysis, keyword matching, resume builder, export to PDF/DOCX).
- Define non-functional requirements (scalability, user-friendliness, data security).

2. Data Collection and Preprocessing

- Collect sample resumes and job descriptions for training and testing.
- Clean and preprocess text by removing stop words, punctuation, and performing tokenization and lemmatization.
- Convert textual data into numerical representations using TF-IDF, Bag-of-Words, or Transformer embeddings (BERT).

3. Resume Analyzer Design

- Implement NLP pipeline to extract structured information from resumes (Education, Experience, Skills, Certifications).
- Apply cosine similarity to compare resume vectors with job description vectors for relevance scoring.
- Generate a scorecard showing:
    - Resume–Job Match Percentage
    - ATS compliance
    - Missing/extra keywords
    - Grammar and formatting suggestions

4. Resume Builder Design

- Develop an interactive web form where users input details (Personal Info, Skills, Experience, Education).
- Integrate AI-assisted text generation to suggest professional summaries, skill phrasing, and impactful content.
- Offer customizable, ATS-friendly resume templates with live preview.
- Provide option to export in PDF/DOCX format.

5. AI/NLP Model Integration

- Use NLP libraries (spaCy, NLTK) for text parsing.
- Integrate pre-trained transformer models (BERT/GPT APIs) for semantic similarity and wording improvements.
- Implement Softmax and similarity scoring functions for ranking resumes against job descriptions.

6. Web Application Development

- Frontend: React/Next.js + Tailwind CSS for responsive UI.
- Backend: Node.js (Express) or Django/Flask for handling requests and AI logic.
- Database: MongoDB/PostgreSQL for storing user data, resumes, and templates.
- Cloud Storage: AWS S3 or Firebase for uploaded resume files.

7. Testing and Evaluation

- Perform functional testing for each module (Analyzer, Builder, Export).
- Conduct accuracy testing of resume scoring against benchmark datasets.
- Collect feedback from users to improve UI/UX and scoring logic.

8. Deployment and Maintenance

- Deploy the web app on cloud platforms (Heroku, AWS, or Vercel).
- Provide regular updates to add new templates and enhance AI models.
- Maintain security for user data through encryption and authentication mechanisms.

System Requirements:

1. Hardware Requirements:

For Development (Developers' System):

- **Processor**: Intel Core i5/i7 (or AMD equivalent) – 2.4 GHz or higher
- **RAM**: Minimum 8 GB (16 GB recommended for AI/NLP tasks)
- **Storage**: 500 GB HDD / 256 GB SSD (preferred for faster data processing)
- **Graphics**: Dedicated GPU (NVIDIA GTX/RTX series) if running heavy AI/ML locally
- **Network**: Stable Internet connection (for API calls, cloud deployment, and database access)

2. Software Requirements:

For Development Environment (Backend + Frontend + AI Integration):

- **Operating System**: Windows 10/11, Linux (Ubuntu), or macOS
- **Programming Languages**:
  - **Python** (for AI/NLP model development – spaCy, NLTK, Transformers)
  - **JavaScript/TypeScript** (for frontend & backend with React/Next.js and Node.js)
- **Frameworks / Libraries**:
  - **Frontend**: React.js / Next.js, Tailwind CSS / Bootstrap
  - **Backend**: Node.js (Express) or Django/Flask (Python-based)
  - **AI/NLP**: spaCy, NLTK, scikit-learn, TensorFlow / PyTorch, Hugging Face Transformers
  - **Text Processing**: PDFPlumber, docx2txt (for resume parsing)
- **Database**: MongoDB / PostgreSQL / MySQL
- **Web Server**: Nginx / Apache (for hosting)
- **Package Managers**: npm / pip
- **Cloud Services (Optional)**: AWS / Firebase / Heroku / Vercel for deployment
- **Version Control**: Git & GitHub/GitLab for collaboration

For End Users (Web App Usage):

- **Operating System**: Any OS (Windows, Linux, macOS, Android, iOS)
- **Web Browser**: Chrome, Firefox, Edge, or Safari (latest versions)
- **PDF/DOCX Viewer**: Required to open downloaded resumes

Data Management

Efficient data management is crucial for handling resume files, job descriptions, AI-processed data, and user profiles. The project incorporates structured storage, preprocessing, and retrieval mechanisms to ensure scalability, security, and high performance.

1. Types of Data Managed

- **User Data**: Profile information (name, email, login credentials).
- **Resume Data**: Uploaded files (PDF/DOCX), extracted text, and AI-parsed sections (Education, Skills, Experience).
- **Job Description Data**: Text data provided by users for matching and scoring.
- **AI Output Data**: Resume scores, keyword match results, missing skills, and recommendations.
- **Template Data**: Predefined resume templates for the Resume Builder module.

2. Storage Mechanism

- **Database**:
  - **MongoDB** (NoSQL) for flexible storage of semi-structured resume/job description data.
  - **PostgreSQL/MySQL** (Relational DB) for structured user accounts, templates, and analytics.
- **File Storage**:
  - Uploaded resumes stored securely in cloud storage (e.g., AWS S3, Firebase Storage).
  - Only extracted text is processed for AI tasks to ensure efficiency.

3. Data Processing

- Resumes and job descriptions are preprocessed using NLP techniques (tokenization, lemmatization, stop-word removal).
- Features are extracted (skills, experience, education) and converted into **vector representations** using TF-IDF or embeddings.
- AI/NLP models analyze this processed data for scoring and recommendations.

4. Data Security & Privacy

- Sensitive data (user details, resumes) is encrypted during storage and transmission (AES-256, HTTPS/SSL).
- Authentication and authorization mechanisms (JWT tokens, OAuth) secure access.
- User resumes are processed temporarily; long-term storage is optional with user consent.

5. Backup and Recovery

- Regular database backups scheduled to prevent data loss.
- Cloud-based storage ensures redundancy and high availability.

6. Data Lifecycle

1. **Collection** → Resume/Job description uploaded by the user.
2. **Storage** → Saved in database + cloud storage.
3. **Processing** → NLP & AI algorithms analyze and extract key features.
4. **Output** → Resume score, recommendations, and generated templates.
5. **Retention/Deletion** → User can save resume data for future use or request permanent deletion

9. **Review/Literature survey of minimum 10 Conference/Journal Papers supporting Project idea**

Sr.No.	Reference (Authors, Year, Venue)	Key contribution	Relevance to project
1	Rosenberger J., *CareerBERT: Matching resumes to ESCO jobs in a shared embedding space*, 2025, *Elsevier/ScienceDirect*. (ScienceDirect)	Proposes CareerBERT — a BERT-based shared embedding approach that transforms resumes and job descriptions into a common semantic space for robust matching beyond simple keywords.	Directly supports using transformer embeddings for semantic resume–JD matching and improves match quality vs. keyword- only methods.
2	Bevara R.V.K., *Resume2Vec: Transforming Applicant Tracking Systems*, 2025, *MDPI Electronics*. (MDPI)	Introduces Resume2Vec, an embedding technique tailored for resume↔JD retrieval; shows better alignment with human judgments than classic ATS heuristics.	Useful for building an embedding-based scoring layer and improving ATS-compatibility metrics.
3	(ArXiv) *AI Hiring with LLMs: A Context-Aware and Explainable Framework*, 2025. (arXiv)	Explores use of large language models (LLMs)for contextual reasoning in hiring—inferring transferable skills and providing explainable evaluations.	Motivates use of LLMs (or APIs) for suggestion generation, contextual skill inference, and explainability in recommendations.

4	Thatavarthi G.S., Sai S.K., *Automated Resume Parsing: A Natural Language Processing Approach*, 2023, IEEE CSITSS (conference). (ResearchGate)	Describes an end-to- end resume parsing pipeline (section detection, entity extraction) using classical NLP and demonstrates extraction accuracy on resume datasets.	Direct blueprint for the resume parsing module (sectionization, entity extraction) in your project.
5	*Enhanced Resume Screening for Smart Hiring Using S-BERT* (The Science and Information Association / TheSAI), 2024/2025. (The Science and Information Organization)	Applies Sentence- BERT (S-BERT) for semantic matching of resumes and job descriptions, showing improved screening precision and reduced false positives.	Supports using sentence- level transformer models to improve semantic matching and ranking in analyzer module.
6	Stanford CS224n project/report: *Document Matching for Job Descriptions* (course report). (Stanford University)	Practical project demonstrating document-embedding + cosine similarity approach to match job descriptions with standardized role descriptions.	Good pedagogical/implementation reference for embedding + cosine-similarity workflow in scoring.
7	IOSR Journals: *Comparison Of Models For Resume–JD Matching: BERT* (2025). (IOSR Journals)	Empirical comparison of BERT vs. TF-IDF / traditional methods for resume–job similarity; BERT outperforms keyword approaches on semantic matches.	Provides experimental evidence to prioritize transformer embeddings for matching.
8	IJESAT / IJ papers: *A Resume Screening Operation Powered by AI / Machine Learning- Based Resume Shortlisting* (2024–2025). (IJESAT)	Multiple recent applied papers describing ML pipelines for resume classification, ranking, and shortlisting with practical datasets and system architectures.	Offers practical system design choices (feature extraction, ranking heuristics) and evaluation setups you can adapt.

9	IJFMR / other journals: *NLP-Powered Resume Matching for Recruitment* (2024). (IJFMR	Demonstrates a pipeline combining TF-IDF, cosine similarity, and ML classifiers to predict binary job-fit labels; includes dataset	Useful for baseline model implementation and evaluation metrics (accuracy, precision, recall) for the Analyzer.
10	TechRxiv / preprints: *Machine learning for recruitment—Analyzing job matching algorithms* (2023–2024 preprints). (TechRxiv)	Survey/experimental study comparing BERT-based pipelines with lighter-weight alternatives; discusses runtime/throughput trade-offs and practical deployment considerations.	Helps choose trade-offs (model size vs. latency) and informs decisions on on- prem vs. cloud inference for production.

10. Plan of Project Execution

Start date From Group Registration till end of project work (30 July 2025 To 17 March 2026).

Activities	Month I (July)	Month II (Aug)	Month III (Sept)	Month IV (Oct)	Month V (Nov)	Month VI (Dec)	Month VII (Jan)	Month VIII (Feb)	Month IX (March)
Start date from Group Registration									
Submission of Synopsis									
Submission of Survey Paper									
Submission of Design Paper									
Last date to submit Report SEM I Spiral, Log Book and Workbook									
Submission of Result Paper with 100% coding									
Technical Event & IPR Activities Certification									
Last date to submit Black Golden Embossing Report SEM II									
Final Viva / Execution of Model									

Table 2: Project Execution Timeline

Prof. Dr. Bere S.S
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